

An SPT-Filtered Euclidean TT Model and Its Conditional Perturbative Properties

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Fifth edition
July 2026

Abstract

This paper isolates the mathematically valid part of a proposed spectral proper-time (SPT) modification of a transverse-traceless graviton propagator. It studies a Euclidean model in which selected internal lines carry a heat-kernel filter. A positive lower endpoint in the filter's proper-time measure gives Gaussian decay, but that endpoint is an additional source or schedule assumption: it does not follow from a positive internal spectral gap. For a fixed graph, the filters suppress every ultraviolet loop direction exactly when the momenta of the filtered lines have full rank on the graph's loop space. Equivalently, the subgraph made only from unfiltered lines must contain no cycle. Thus one filtered graviton line is not enough in a general multiloop graph. We also prove that a nonzero propagator with a positive Kallen–Lehmann or Stieltjes representation cannot have permanent Gaussian decay in the same Euclidean spectral variable. The earlier claims of automatic reflection positivity, causal support, all-loop finiteness, BRST consistency, and unitarity are therefore withdrawn. The construction remains useful as a conditional Euclidean filter or regulator model. It is not, by itself, a physical ultraviolet completion of quantum gravity or a selected consequence of Modal Triplet Theory.

Revision note for this edition

Supersedes. *Modal Triplet Theory: From MTT to a UV-Finite, Unitary Quantum Gravity*, version 4.

Reason. Version 4 inferred an external Gaussian filter from an internal gap, converted a positive Laplace representation into a Stieltjes representation without a valid inversion theorem, and claimed that one filtered graviton line made every mixed graph ultraviolet finite. Those steps are false in the stated generality.

Resolution. This edition retitles and narrows the work to a Euclidean TT model. It separates the spectral-gap and proper-time-gap assumptions, proves the exact loop-rank criterion, supplies an explicit two-loop failure of the earlier argument, and proves the incompatibility between a nonzero positive spectral propagator and permanent Gaussian decay.

Retained result. A proper-time support gap does give a precise heat-kernel factorization and strong Euclidean graph bounds whenever the filtered momenta span the full loop space.

Remaining boundary. No current theorem selects this filter from the physical q79 MTT geometry or proves reflection positivity, Lorentzian causal support, gauge consistency, unitarity, or ultraviolet completion for the filtered theory.

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1 Purpose and Orientation

The original proposal combined three logically different objects:

1. an internal positive operator with a spectral gap;
2. a Euclidean heat-kernel multiplier acting on external momentum; and
3. a Lorentzian gauge theory intended to describe physical gravitons.

Each object is useful, but none automatically supplies the next. A gap in an internal finite or compact operator controls internal inversion. A heat-kernel multiplier controls Euclidean high-frequency behavior. A physical quantum-gravity model must additionally establish a Lorentzian state space, gauge identities, causal propagation, and a positive physical spectrum.

This paper therefore asks a smaller and answerable question:

Given an explicitly imposed Euclidean filter, exactly which fixed Feynman graphs does it control, and which physical conclusions do not follow?

The answer is useful both positively and negatively. It gives a clean criterion for filtered calculations and prevents a regulator estimate from being mistaken for a unitarity theorem.

In plain language, a filter placed on one pipe in a multiloop network does not slow every independent circulation through that network. It succeeds only when the filtered pipes intercept the entire cycle space. Likewise, a positive Euclidean smoothing rule is not

automatically a positive spectrum of physical particles. The two main theorems below make those distinctions exact.

2 The Euclidean TT Filter

2.1 Model propagator

Work first in a Euclidean momentum chart. Away from zero momentum, let $\Pi_{\text{TT}}(p)$ denote the usual transverse-traceless tensor projector and consider

$$\Delta_0(p) = \frac{\Pi_{\text{TT}}(p)}{p^2 + m_{\text{IR}}^2}. \quad (2.1)$$

The nonnegative m_{IR} is only an infrared regulator. It is not a claim that the physical graviton is massive. Let $f : [0, \infty) \rightarrow \mathbb{R}$ be a bounded filter and define

$$\Delta_f(p) = f(p^2) \Delta_0(p). \quad (2.2)$$

The Gaussian example is $f(x) = e^{-\tau_0 x}$, with $\tau_0 > 0$.

Equation (2.2) is a model definition. To derive it from MTT, one would need a selected source map whose compression of the physical operator equals this multiplier while preserving the required gauge and causal structures. No such theorem is assumed here.

2.2 Proper-time support

Definition 2.1 (SPT support-gap hypothesis). The filter satisfies the SPT support-gap hypothesis if there is a finite positive measure μ and a number $\tau_0 > 0$ such that

$$f(x) = \int_{[\tau_0, \infty)} e^{-tx} \mu(dt). \quad (2.3)$$

Proposition 2.2 (Exact SPT factorization). *Let $L \geq 0$ be self-adjoint and let f satisfy Definition 2.1. Then*

$$f(L) = e^{-\tau_0 L/2} f_0(L) e^{-\tau_0 L/2}, \quad f_0(x) = \int_{[\tau_0, \infty)} e^{-(t-\tau_0)x} \mu(dt), \quad (2.4)$$

where $f_0(L)$ is bounded and positive. Moreover,

$$0 \leq f(x) \leq \mu([\tau_0, \infty)) e^{-\tau_0 x}. \quad (2.5)$$

Proof. The spectral functional calculus permits the factor $e^{-\tau_0 L}$ to be removed from each integrand in (2.3). Splitting that factor symmetrically gives (2.4). Positivity follows from positivity of μ , and (2.5) follows because $t \geq \tau_0$. $\square$

The proposition is exact, but its premise matters. A discrete update schedule can be chosen so that its Laplace measure is supported on $\{n\tau_0 : n \geq 1\}$. That construction proves the implication

$$\text{declared schedule} \implies \text{proper-time support gap}.$$

It does not prove that the physical MTT source selects that schedule.

3 An Internal Gap Does Not Produce External Damping

The earlier paper treated an internal spectral gap and an external proper-time support gap as though they were the same statement. A simple compression shows why they are not.

Theorem 3.1 (Gap-to-filter implication fails). *Let A_{int} be positive self-adjoint on $\mathcal{H}_{\text{int}}$, and suppose it has a normalized eigenvector e_0 with eigenvalue $\lambda_* > 0$. For the external spectral variable $x \geq 0$, define*

$$A(x) = xI + A_{\text{int}}, \quad Bu = \langle e_0, u \rangle.$$

Then the compressed resolvent is

$$BA(x)^{-1}B^* = \frac{1}{x + \lambda_*}. \quad (3.1)$$

It has only algebraic decay. Consequently, a positive internal gap alone does not imply an external factor $e^{-\tau_0 x}$ with $\tau_0 > 0$.

Proof. Since $A(x)e_0 = (x + \lambda_*)e_0$, compression to the span of e_0 gives (3.1). For every $C, \tau_0 > 0$,

$$\frac{1}{x + \lambda_*} \not\leq C \frac{e^{-\tau_0 x}}{x + \lambda_*}$$

for all sufficiently large x . Thus the gap does not supply the claimed Gaussian factor. $\square$

The internal gap can still be valuable: it controls inverse norms and decoupling errors. What it cannot do without another theorem is determine the ultraviolet dependence of the external momentum variable.

4 Exactly Which Graphs Are Gaussian-Controlled

4.1 Loop-momentum map

Let a connected graph Γ have L independent loop momenta $\ell = (\ell_1, \dots, \ell_L) \in (\mathbb{R}^d)^L$. For every filtered internal line j , write

$$k_j(\ell, Q) = \sum_{a=1}^L C_{ja} \ell_a + q_j(Q), \quad (4.1)$$

where Q collects external momenta. If line j has Gaussian parameter $\tau_j > 0$, the combined damping is

$$\exp \left(- \sum_j \tau_j |k_j(\ell, Q)|^2 \right).$$

The scalar matrix $C_\Gamma = (C_{ja})$ records how the filtered lines see the loop space.

Theorem 4.1 (Filtered-loop rank criterion). *Assume Euclidean propagator denominators are infrared regulated and vertices grow at most polynomially in loop momenta. The Gaussian factors suppress every ultraviolet loop direction if and only if*

$$\text{rank } C_\Gamma = L. \quad (4.2)$$

When (4.2) holds, the fixed graph integral is absolutely ultraviolet convergent. When it fails, the filters alone provide no decay along at least one nonzero loop direction; ordinary propagator power counting and renormalization must decide that direction.

Proof. Let $W = \text{diag}(\tau_j)$. If C_Γ has full column rank, $C_\Gamma^T W C_\Gamma$ is positive definite. For fixed Q , completing the square, or using $|a + b|^2 \geq \frac{1}{2}|a|^2 - |b|^2$, gives constants $c > 0$ and $K_Q < \infty$ such that

$$\sum_j \tau_j |k_j(\ell, Q)|^2 \geq c|\ell|^2 - K_Q.$$

Hence the integrand is bounded by a polynomial times $e^{-c|\ell|^2}$, which is integrable on $\mathbb{R}^{dL}$.

If the rank is smaller, choose $0 \neq v \in \ker C_\Gamma$. Along $\ell = \ell_0 + tv$, every filtered line momentum is independent of t , so the Gaussian contributes no decay as $|t| \rightarrow \infty$. This does not force divergence, but it invalidates any convergence proof based only on the filters. $\square$

Corollary 4.2 (Cycle formulation). *Let D be the set of filtered edges and U the set of unfiltered edges. For the standard graph momentum routing, C_Γ has full loop rank if and only if the subgraph spanned by U is a forest.*

Proof. The loop space is the cycle space of Γ . The map C_Γ is its restriction to the filtered edges. Its kernel consists precisely of circulations supported entirely on U . Such a nonzero circulation exists exactly when the unfiltered subgraph contains a cycle. $\square$

This formulation makes the earlier error visible. “At least one filtered line in the graph” is much weaker than “no loop can be routed entirely through unfiltered lines.”

4.2 Two worked examples

Example 4.3 (One loop). For one loop momentum k , a filtered line carrying $k + Q$ gives $C_\Gamma = (1)$. The rank is one, so the Gaussian controls the single ultraviolet direction.

Example 4.4 (Two-loop failure and repair). Let the independent loop momenta be k and q . If only a line carrying k is filtered, then

$$C_\Gamma = \begin{pmatrix} 1 & 0 \end{pmatrix}, \quad \text{rank } C_\Gamma = 1 < 2.$$

Along $k = 0, q \rightarrow \infty$, the filter is constant. This is an explicit counterexample to the version 4 claim.

If lines carrying k and $k + q$ are both filtered, then

$$C_\Gamma = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}, \quad \det C_\Gamma = 1.$$

Both loop directions are then Gaussian-controlled.

For mixed gravity–matter diagrams, Corollary 4.2 says that every cycle made solely from unfiltered matter lines remains outside the SPT argument. Such subgraphs require the usual BPHZ, Epstein–Glaser, or EFT renormalization analysis [1, 2, 3].

5 Laplace Positivity Is Not Spectral Positivity

5.1 Two different integral representations

A completely monotone Euclidean function has a positive Laplace representation,

$$F(x) = \int_0^\infty e^{-tx} \mu(dt). \tag{5.1}$$

A Stieltjes two-point function has the more restrictive form

$$D(x) = \int_0^\infty \frac{\rho(ds)}{x+s}, \quad \rho \geq 0. \quad (5.2)$$

The second form is the Euclidean counterpart of a positive Kallen–Lehmann spectral representation for a scalar physical component [4, 5]. Every Stieltjes function is completely monotone, but the converse is false [6, 7].

The elementary function e^{-x} already proves strictness: it is the Laplace transform of a point mass at $t = 1$, yet it cannot have the Stieltjes form (5.2) with nonzero positive measure.

Theorem 5.1 (Positive spectral tail excludes permanent Gaussian decay). *Let*

$$D(x) = \int_0^\infty \frac{\rho(ds)}{x+s}, \quad x > 0,$$

for a nonzero positive measure ρ . Then there do not exist $C, \tau, \lambda > 0$ such that

$$D(x) \leq C \frac{e^{-\tau x}}{x + \lambda} \quad (5.3)$$

for all sufficiently large x .

Proof. Choose $R < \infty$ for which $m = \rho([0, R]) > 0$. Positivity gives

$$D(x) \geq \frac{m}{x + R}.$$

Combining this with (5.3) would imply

$$\frac{m}{C} \leq e^{-\tau x} \frac{x + R}{x + \lambda} \rightarrow 0,$$

which is impossible. □

Remark 5.2 (Scope of the no-go). The theorem does not prove that every nonlocal quantum theory is nonunitary. It proves that one specific argument is unavailable: a permanently Gaussian-damped nonzero propagator cannot obtain positive physical spectral weight from the same Stieltjes representation. Gauge theories and tensor fields require additional attention to constraints and physical cohomology, making it even more important not to infer unitarity from the scalar Laplace formula alone.

Osterwalder–Schrader reconstruction requires reflection positivity together with the other Euclidean axioms [8, 9]. Positivity of the heat kernel in (5.1) is not a substitute for that condition.

6 What the Filter Does Not Prove

6.1 Causal support

Euclidean rapid decay is a statement about large positive covectors. A Lorentzian retarded kernel requires a hyperbolic operator and a specified boundary-value prescription. Analytic continuation of an entire Gaussian multiplier can grow in timelike directions, and nonlocal operators need not have light-cone-supported retarded kernels. Causal support must therefore be proved independently; it does not follow from Proposition 2.2.

6.2 BRST and gauge consistency

The TT projector describes the free physical tensor sector after constraints have been handled. In an interacting theory, modifying selected propagator lines after gauge fixing need not preserve the Slavnov–Taylor identities or the quantum master equation. A legitimate filtered gauge theory would need one of the following:

- a gauge-invariant or BV-compatible filtered action;
- a regulator construction with controlled symmetry breaking and a proved regulator-removal limit; or
- an independent physical-state construction in which the filtered observables act on positive BRST cohomology.

Standard BV and perturbative algebraic QFT results remain available for the local effective theory under their usual hypotheses [10, 11, 12, 13, 14, 15]. Importing those results does not establish them for the permanent SPT modification.

6.3 Matter subgraphs and all-orders language

If the unfiltered subgraph contains a loop, the SPT filter does not control that loop. Pure matter subgraphs and rank-deficient mixed subgraphs must be renormalized by standard local methods. Even when every fixed graph is absolutely convergent, this does not by itself establish convergence of the sum over graphs, nonperturbative existence, reflection positivity, or a unitary Lorentzian scattering operator.

7 The Infrared Statement That Does Survive

Suppose f is differentiable at zero and normalized by $f(0) = 1$. Then

$$f(p^2) = 1 + f'(0)p^2 + O(p^4),$$

so

$$\Delta_f(p) = \Delta_0(p) \left(1 + f'(0)p^2 + O(p^4) \right). \quad (7.1)$$

For $f(p^2) = e^{-\tau_0 p^2}$, the first correction is $-\tau_0 p^2$. Thus the filter can be chosen to preserve the free massless-pole residue in a low-momentum expansion.

Equation (7.1) is only an infrared matching statement for the model propagator. Recovering General Relativity also requires the nonlinear vertices, universal conserved stress coupling, gauge identities, and physical Newton normalization. None follows from $f(0) = 1$.

8 Placement Inside the Current MTT Program

The current MTT corpus contains an exact finite internal transverse-traceless support result on its selected helicity-two carrier. It also contains a conditional coherent reduction to the four-dimensional Einstein action. Those are distinct from the SPT filter studied here:

Object	Current status	What it does not yet provide
Finite internal TT support	exact on the declared finite carrier	Lorentzian graviton pole, Newton normalization, or full stress response
Controlled coherent GR reduction	conditional on a higher-dimensional Einstein sector, truncation, and gap assumptions	derivation of those inputs from the upper MTT source
SPT proper-time endpoint τ_0	model assumption or declared schedule	selection by q79 geometry or identification with a physical Planck scale
Filtered graph bounds	exact under Theorem 4.1	reflection positivity, BRST consistency, causality, or a sum over graphs

In particular, a dimensionless internal eigenvalue such as the selected TT value recorded in the finite carrier is not automatically a physical mass or cutoff. A units-and-normalization map is required before it can be compared with the Planck scale. The broader paper *Perturbative Coherent-Sector Quantum Gravity and the Heterotic UV-Completion Boundary* should use the present paper only for the SPT model and its no-go results; its physical ultraviolet discussion belongs to the separate q79 heterotic worldsheet contract.

9 Claim Ledger and Conclusion

Claim	Verdict	Reason
SPT support gap implies Gaussian Euclidean damping	proved	Proposition 2.2
Internal spectral gap implies an SPT support gap	false in general	Theorem 3.1
One filtered graviton line makes every graph finite	withdrawn	Two-loop rank counterexample
Full filtered-loop rank makes a fixed graph UV convergent	proved	Theorem 4.1
Positive Laplace kernel implies positive spectral density	false in general	Laplace and Stieltjes classes differ
Positive spectral density coexists with permanent Gaussian decay	impossible for a nonzero propagator	Theorem 5.1
SPT alone proves BRST, causality, OS positivity, or unitarity	not proved	Each requires an independent construction
SPT is the selected MTT ultraviolet completion	not established	No same-source selection theorem

The corrected conclusion is narrower but firmer. SPT is a useful Euclidean filter model with an exact support-gap factorization and an exact graph-rank criterion. It can regularize selected calculations and expose which loop directions remain uncontrolled. It does not turn

an internal MTT gap into a physical ultraviolet completion. The route from MTT geometry to quantum gravity must instead supply a selected Lorentzian operator or worldsheet theory together with its gauge, positivity, causal, normalization, and nonperturbative contracts.

Computational Evidence and Reproducibility

The SPT-filtered transverse-traceless model has the perturbative properties proved in this paper only. The open strict-upgrade ledger does not establish reflection positivity, nonperturbative unitarity, or ultraviolet-complete quantum gravity.

The referenced rows are frozen to the curated results repository state identified below. Hashes are grouped in eight-character blocks for line breaking.

Repository: <https://github.com/PeterNero/mtt-results-repro>
Commit: 31247ebb 5c22f3fb b5443024 365433c6 ee0bff4a
Manifest: release/result_manifest.json
Manifest SHA-256:
fb399689 60b00584 631dbf53 1a708e18
ef928d6b 6d935119 c185d7f6 32b1e7cd

Tier labels are quoted verbatim from that manifest. A row used directly supports only the specific computational statement identified above; a corpus-state cross-check does not prove this paper’s local theorems; and an open row is evidence of an unresolved obligation, never of closure.

Open boundary (not evidence of closure).

- A05/strict_upgrade_ledger (*open*).
Current 2/9 strict no-knob upgrade ledger.

No imported row changes theorem ownership or promotes a neighboring claim: all local statements retain their stated hypotheses, domains, and limitations.

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